

CSE 150A-250A AI: Probabilistic Methods

Lecture 3

Fall 2026

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Slides adapted from previous versions of the course (Prof. Lawrence, Prof. Alvarado, Prof Berg-Kirkpatrick)

Agenda

Review

Joint Distributions and Inference

Alarm Example

Belief networks

Review

- Types of probabilities:

$P(X, Y)$ joint

$P(Y|X)$ conditional

$P(X)$ unconditional (or marginal)

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- Types of probabilities:

$P(X, Y)$ joint

$P(Y|X)$ conditional

$P(X)$ unconditional (or marginal)

- Useful rules:

$P(A, B, C, \dots) = P(A) P(B|A) P(C|A, B) \dots$ product rule

$P(X|Y) = P(Y|X)P(X)/P(Y)$ Bayes rule

$P(X) = \sum_y P(X, Y=y)$ marginalization

Review

- Types of probabilities:

$P(X, Y)$ joint

$P(Y|X)$ conditional

$P(X)$ unconditional (or marginal)

- Useful rules:

$$P(A, B, C, \dots) = P(A) P(B|A) P(C|A, B) \dots \quad \text{product rule}$$

$$P(X|Y) = P(Y|X)P(X)/P(Y) \quad \text{Bayes rule}$$

$$\underline{P(X)} = \sum_y \underline{P(X, Y=y)} \quad \text{marginalization}$$

- Conditioning on background evidence E :

$$P(A, B, C, \dots | E) = P(A|E) P(B|A, E) P(C|A, B, E) \dots$$

$$P(X|Y, E) = P(Y|X, E)P(X|E)/P(Y|E)$$

$$P(X|E) = \sum_y P(X, Y=y|E)$$

Marginal and conditional independence

- Marginal independence

$$\begin{array}{l} P(X|Y) = P(X) \\ P(Y|X) = P(Y) \\ P(X, Y) = P(X) P(Y) \end{array} \left. \vphantom{\begin{array}{l} P(X|Y) = P(X) \\ P(Y|X) = P(Y) \\ P(X, Y) = P(X) P(Y) \end{array}} \right\} \begin{array}{l} \text{Each of these} \\ \text{implies the} \\ \text{other two.} \end{array}$$

$$P(X) P(Y|X)$$

$$P(X) P(Y)$$

Marginal and conditional independence

- Marginal independence

$$P(X|Y) = P(X)$$

$$P(Y|X) = P(Y)$$

$$P(X, Y) = P(X)P(Y)$$

*Each of these
implies the
other two.*

- Conditional independence

$$P(X|Y, E) = P(X|E)$$

$$P(Y|X, E) = P(Y|E)$$

$$P(X, Y|E) = P(X|E)P(Y|E)$$

*Each of these
also implies the
other two.*

Example of conditional dependence



- B and E are marginally independent:

$$\underline{\underline{P(B)}} = \underline{\underline{P(B|E)}}$$

$$\underline{\underline{P(E)}} = \underline{\underline{P(E|B)}}$$

$$\underline{\underline{P(B, E)}} = \underline{\underline{P(B) P(E)}}$$

Example of conditional dependence



- B and E are marginally independent:

$$P(B) = P(B|E)$$

$$P(E) = P(E|B)$$

$$P(B, E) = P(B)P(E)$$

- But B and E are conditionally dependent given A :

$$P(B|A) \neq P(B|E, A)$$

$$P(E|A) \neq P(E|B, A)$$

$$P(B, E|A) \neq P(B|A)P(E|A)$$

Joint Distributions and Inference

Why Joints Matter

The joint distribution $P(X_1, \dots, X_n)$ is a **complete description of uncertainty**.

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From the full joint distribution $P(X_1, \dots, X_n)$, what kinds of probabilities can we compute?

- A. Only marginals, $P(X_i)$
- B. Only conditionals, $P(X_i|X_j)$
- ☒ C. Any marginal or conditional over the variables
- D. None of the above

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Inference: Compute the **posterior** distribution of query variables given observed evidence.

- Model complexity

Suppose $X_i \in \{0, 1\}$ are binary random variables.

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How many numbers do we need to specify the joint distribution of $P(X_1=x_1, \dots, X_n=x_n)$?

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B. $O(2^n)$

C. $O(n^2)$

D. $O(\log n)$

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It requires $O(2^n)$ numbers to specify the joint distribution $P(X_1=x_1, \dots, X_n=x_n)$.

Representation

- **Representation:** compactly encode the joint.

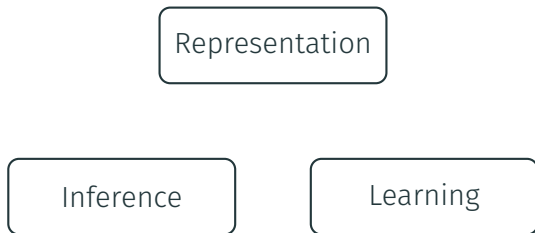
Conceptual and Practical Goals

Representation

Inference

- **Representation:** compactly encode the joint.
- **Inference:** answer queries given evidence.

Conceptual and Practical Goals



- **Representation:** compactly encode the joint.
- **Inference:** answer queries given evidence.
- **Learning:** estimate structure/parameters from data.

Alarm Example

Alarm example



Alarm example



- Binary random variables

$B \in \{0, 1\}$ Was there a burglary?

$E \in \{0, 1\}$ Was there an earthquake?

$A \in \{0, 1\}$ Was the alarm triggered?

$J \in \{0, 1\}$ Did Jamal call?

$M \in \{0, 1\}$ Did Maya call?

- Product rule

$$P(B, E, A, J, M)$$

- Product rule

$$P(\underline{B}, E, A, J, M)$$

$$= \left(\underline{P(B)} P(E|B) P(A|B,E) P(J|A,B,E) \right. \\ \left. \rightarrow P(M|A,B,E,J) \right)$$

- Product rule

$$P(B, E, A, J, M)$$

$$= P(B) P(E|B)$$

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Joint distribution

- Product rule

$$P(B, E, A, J, M) \stackrel{2^5}{=} P(B) P(E|B) P(A|B, E) P(J|B, E, A) P(M|B, E, A, J)$$

Note: the above is true no matter what the variables signify.

$$P(E|B) = \underline{\underline{P(E)}}$$

- Product rule

$$P(B, E, A, J, M)$$

$$= P(B) P(E|B) P(A|B, E) P(J|B, E, A) P(M|B, E, A, J)$$

Note: the above is true no matter what the variables signify.

- Domain-specific assumptions

- Product rule

$$\begin{aligned} P(B, E, A, J, M) \\ = P(B) P(E|B) P(A|B, E) P(J|B, E, A) P(M|B, E, A, J) \end{aligned}$$

Note: the above is true no matter what the variables signify.

- Domain-specific assumptions

$$P(E|B) = P(E) \quad \text{marginal independence}$$

- Product rule

$$\begin{aligned} P(B, E, A, J, M) \\ = P(B) P(E|B) P(A|B, E) P(J|B, E, A) P(M|B, E, A, J) \end{aligned}$$

Note: the above is true no matter what the variables signify.

- Domain-specific assumptions

$$\begin{array}{ll} P(E|B) = P(E) & \text{marginal independence} \\ P(J|B, E, A) = P(J|A) & \text{conditional independence} \end{array}$$

- Product rule

$$\begin{aligned} P(B, E, A, J, M) \\ = P(B) P(E|B) P(A|B, E) P(J|B, E, A) P(M|B, E, A, J) \end{aligned}$$

Note: the above is true no matter what the variables signify.

- Domain-specific assumptions

$P(E B)$	$=$	$P(E)$	marginal independence
$P(J B, E, A)$	$=$	$P(J A)$	conditional independence
$P(M B, E, A, J)$	$=$	$P(M A)$	conditional independence

Completing the model

- Joint distribution

$$P(B, E, A, J, M)$$

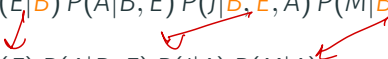
$$= P(B) P(E|B) P(A|B, E) P(J|B, E, A) P(M|B, E, A, J)$$

Completing the model

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Completing the model

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$$= P(B) P(E|B) P(A|B, E) P(J|B, E, A) P(M|B, E, A, J)$$

$$= \underbrace{P(B)} \underbrace{P(E)} \underbrace{P(A|B, E)} \underbrace{P(J|A)} \underbrace{P(M|A)}$$

- Conditional probability tables (CPTs)

Completing the model

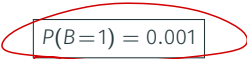
- Joint distribution

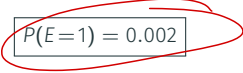
$$P(B, E, A, J, M)$$

$$= P(B) P(E|B) P(A|B, E) P(J|B, E, A) P(M|B, E, A, J)$$

$$= \underbrace{P(B)} \underbrace{P(E)} \underbrace{P(A|B, E)} P(J|A) P(M|A)$$

- Conditional probability tables (CPTs)


$$P(B=1) = 0.001$$


$$P(E=1) = 0.002$$

Completing the model

- Joint distribution

$$P(B, E, A, J, M)$$

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- Conditional probability tables (CPTs)

$$P(B=1) = 0.001$$

$$P(E=1) = 0.002$$

B	E	$P(A=1 B, E)$
0	0	0.001
1	0	0.94
0	1	0.29
1	1	0.95

Completing the model

- Joint distribution

$$P(B, E, A, J, M)$$

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- Conditional probability tables (CPTs)

$P(B=1) = 0.001$

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B	E	$P(A=1 B, E)$
0	0	0.001
1	0	0.94
0	1	0.29
1	1	0.95

A	$P(J=1 A)$
0	0.05
1	0.9

Completing the model

- Joint distribution

$$B \perp E$$

$$P(E|B) = P(E)$$

→

$$\begin{aligned} P(B, E, A, J, M) &= P(B) P(E|B) P(A|B, E) P(J|B, E, A) P(M|B, E, A, J) \\ &= P(B) P(E) P(A|B, E) P(J|A) P(M|A) \end{aligned}$$

- Conditional probability tables (CPTs)

$$P(B=1) = 0.001$$

$$P(E=1) = 0.002$$

B	E	$P(A=1 B, E)$
0	0	0.001
1	0	0.94
0	1	0.29
1	1	0.95

A	$P(J=1 A)$
0	0.05
1	0.9

A	$P(M=1 A)$
0	0.01
1	0.7

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$$P(\underbrace{B=1}, \underbrace{E=0}, \underbrace{A=1}, \underbrace{J=1}, \underbrace{M=1})$$

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- Joint probabilities are easy to compute:

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$$= P(B=1) P(E=0) P(A=1|B=1, E=0) P(J=1|A=1) P(M=1|A=1)$$

$$= (0.001)(1 - 0.002)(0.94)(0.9)(0.7)$$

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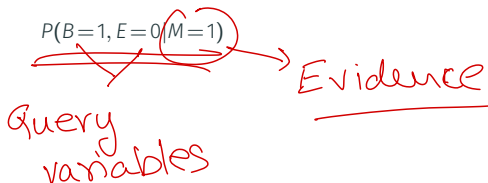
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- Any inference can be expressed in terms of joint probabilities:



A handwritten diagram illustrating the components of a probability expression. The expression $P(B=1, E=0, M=1)$ is shown. The term $M=1$ is circled in red. A red arrow points from the circle to the word "Evidence", which is underlined in red. A red line is drawn under the entire expression $P(B=1, E=0, M=1)$. The words "Query variables" are written in red below the expression.

$$P(B=1, E=0, M=1)$$

Query variables

Evidence

- Joint probabilities are easy to compute:

$$\begin{aligned} & P(B=1, E=0, A=1, J=1, M=1) \\ &= P(B=1) P(E=0) P(A=1|B=1, E=0) P(J=1|A=1) P(M=1|A=1) \\ &= (0.001)(1 - 0.002)(0.94)(0.9)(0.7) \end{aligned}$$

- Any inference can be expressed in terms of joint probabilities:

$$P(B=1, E=0|M=1)$$

$$\sum_{a_j} \frac{P(B=1, E=0, M=1)}{P(M=1)}$$

product rule

- Joint probabilities are easy to compute:

$$\begin{aligned} &P(B=1, E=0, A=1, J=1, M=1) \\ &= P(B=1) P(E=0) P(A=1|B=1, E=0) P(J=1|A=1) P(M=1|A=1) \\ &= (0.001)(1 - 0.002)(0.94)(0.9)(0.7) \end{aligned}$$

- Any inference can be expressed in terms of joint probabilities:

$$\begin{aligned} &P(B=1, E=0|M=1) \\ &= \frac{P(B=1, E=0, M=1)}{P(M=1)} \quad \boxed{\text{product rule}} \\ &= \frac{\sum_{a,j} P(B=1, E=0, A=a, J=j, M=1)}{\sum_{b',e',a',j'} P(B=b', E=e', A=a', J=j', M=1)} \quad \boxed{\text{marginalization}} \end{aligned}$$

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But this approach can be very inefficient!

How to perform inference more efficiently?

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1. Visualize models as directed acyclic graphs.
2. Exploit graph structure to organize and simplify calculations.

Belief networks

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$$\text{BN} = \text{DAG} + \text{CPTs}$$

From distributions to graphs

- It is always true from the product rule that

$$P(X_1, X_2, \dots, X_n) = P(X_1) P(X_2|X_1) \dots P(X_n|X_1, \dots, X_{n-1})$$

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- But suppose in a particular domain that

$$P(X_i | \underbrace{X_1, X_2, \dots, X_{i-1}}_{\text{parents}(X_i)}) = P(X_i | \text{parents}(X_i)),$$

where $\text{parents}(X_i)$ is a subset of $\underbrace{\{X_1, \dots, X_{i-1}\}}$.

From distributions to graphs

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- But suppose in a particular domain that

$$P(X_i|X_1, X_2, \dots, X_{i-1}) = P(X_i|\text{parents}(X_i)),$$

where $\text{parents}(X_i)$ is a subset of $\{X_1, \dots, X_{i-1}\}$.

- **Big idea:** represent conditional dependencies by a DAG.

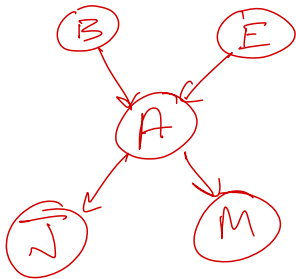
Visualizing the model - Alarm Example

- Joint distribution

$$P(B, E, A, J, M)$$

$$= P(B) P(E|B) P(A|B, E) P(J|B, E, A) P(M|B, E, A, J)$$

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- Directed acyclic graph (DAG)

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Absent edges encode assumptions of independence.

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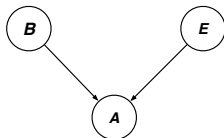
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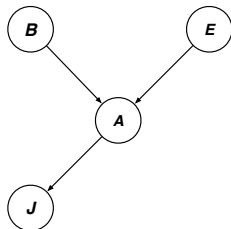
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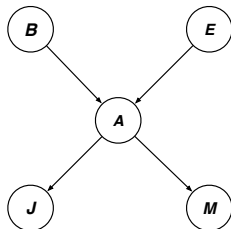
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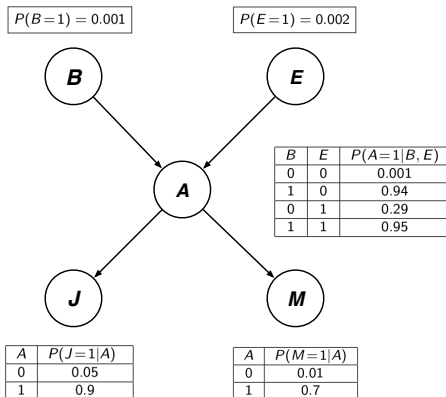
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- Directed acyclic graph (DAG)



Absent edges encode assumptions of independence.

Alarm belief network



This visual representation of the joint distribution is called a **belief network** (or a **Bayesian network**, or a **probabilistic graphical model**).

- Given a BN, the graph and CPTs specify the joint distribution.

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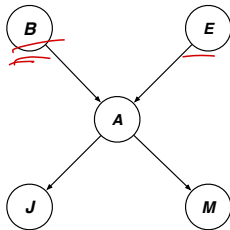
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- To read the factorization from a graph:
 1. write one factor for each node;
 2. condition each node on its parents;
 3. multiply the factors.

Alarm BN:

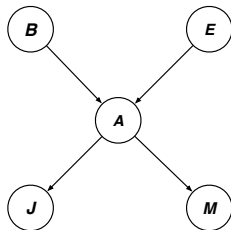


$$P(B) \cdot P(E)$$

$$P(A|B, E)$$

$$P(J|A) \cdot P(M|A)$$

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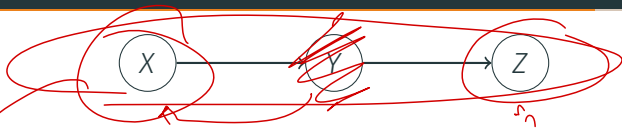


$$P(B, E, A, J, M) = P(B) P(E) P(A|B, E) P(J|A) P(M|A)$$

Example: graph $\rightarrow$ factorization $\rightarrow$ independence



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What independence relationship is implied by this BN?

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$$P(Z|X, Y) = P(Z|Y) \\ (X \perp Z | Y)$$

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Qualitative

DAGs encode assumptions of marginal and conditional independence.

Quantitative

CPTs encode numerical influences of some variables on others.

That's all folks!

Addendum

Constructing BNs and node ordering

Constructing a belief network

Three steps:

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1. Choose your random variables of interest.

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- (c) define the conditional probability table $P(X_i | \text{parents}(X_i))$

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Add the “root causes,” then the variables they influence, then the next variables that are influenced, etc. This often gives a compact graph, but it is not a formal guarantee of optimality.

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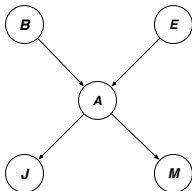
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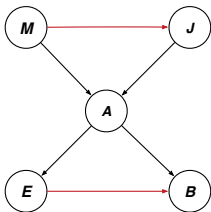
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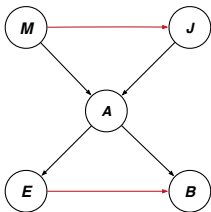
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This belief network has **two extra edges**.

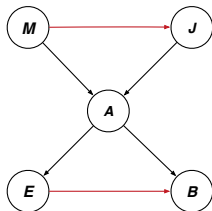
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This DAG does not show $P(B) = P(B|E)$.
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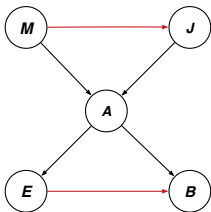
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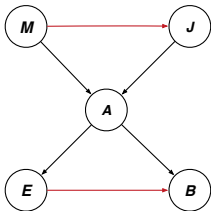
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These CPTs may be more difficult to assess.